

ASSIGNMENT #04

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
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Due Date: Friday, March 20th TA's Mailbox

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Analysis of an Exponential Equation of State.

The following **exponential** Equation of State (EOS) has been proposed as an improvement to the Van der Waals equation.

$$\frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a}{VRT}\right)$$

$$y = \frac{b}{4V} \quad a = \lambda_1 \frac{(RT_c)^2}{P_c} \quad b = \lambda_2 \frac{RT_c}{P_c}$$

Data on the Nitrogen gas compressibility factor were reported by Lunbeck, Michels and Wolkers, *Appl. Sci. Res.*, **A3**: 197 (1952).

P (atm)	Temperature (deg C)				
	0	25	50	75	100
0	1.000	1.000	1.000	1.000	1.000
1	1.000	1.000	1.000	1.000	1.000
3	0.999	0.999	1.000	1.000	1.001
5	0.998	0.999	1.000	1.001	1.001
10	0.996	0.998	1.000	1.001	1.002
30	0.959	0.996	1.001	1.005	1.007
50	0.958	0.996	1.004	1.009	1.014
100	0.984	1.004	1.018	1.027	1.033
200	1.036	1.057	1.072	1.082	1.089
300	1.132	1.146	1.154	1.160	1.163
400	1.256	1.254	1.253	1.250	1.247
600	1.524	1.495	1.471	1.450	1.432
800	1.798	1.723	1.697	1.658	1.625
1000	2.067	1.988	1.923	1.867	1.819

1. Using the proper thermodynamic and mathematical arguments estimate the numerical values of the constants $\{\lambda_1, \lambda_2\}$ and the critical compressibility factor.
2. Perform a nonlinear regression and calculate the constants $\{\lambda_1, \lambda_2\}$ using the experimental data for Nitrogen given in the table above.
3. Comment with a brief paragraph on whether the values calculated from the nonlinear regression are thermodynamically consistent
4. Derive all the formulas for the calculation of the primary residual thermal properties, fugacity coefficient, enthalpy and entropy.
5. Derive all the formulas for the calculation of the secondary thermal properties in the following order: thermal expansion coefficient, heat capacity under constant volume, heat capacity under constant pressure, Joule-Thomson coefficient and speed of sound.

6. Calculate at temperature 10 °C and pressure 830 bar

1. The molar enthalpy in **kJ/mol**
2. The molar entropy in **kJ/(kmol K)**
3. The thermal expansion coefficient in **K⁻¹**
4. The molar heat capacity C_V in **kJ/(kmol K)**
5. The molar heat capacity C_p in **kJ/(kmol K)**
6. The Joule-Thomson coefficient in **K/bar**
7. The speed of sound in **m/s**

7. Prepare a Table with the following columns

- a. Column 1: Property description (See Excel Template **Assignment-04.xlsm**)
- b. Column 3: Value of property from **Assignment #03**
- c. Column 4: Value of property using the Peng-Robinson Equation of State
- d. Column 5: Value of property using the exponential EOS with theoretical values of $\{\lambda_1, \lambda_2\}$
- e. Column 6: Value of property using the exponential EOS with numerical values of $\{\lambda_1, \lambda_2\}$
- f. Column 7: NIST value from the table below

NIST Data

Isothermal Data for T = 10.000 C										
Temperature (C)	Pressure (bar)	Density (mol/l)	Volume (l/mol)	Internal Energy (kJ/mol)	Enthalpy (kJ/mol)	Entropy (J/mol*K)	Cv (J/mol*K)	Cp (J/mol*K)	Sound Spd. (m/s)	Joule-Thomson (K/bar)
10.000	800.00	19.302	0.051807	6.6840	10.829	48.874	23.750	38.775	764.43	-0.046582
10.000	830.00	19.596	0.051032	6.6479	10.884	48.523	23.821	38.731	778.50	-0.047978
10.000	860.00	19.877	0.050308	6.6135	10.940	48.186	23.891	38.691	792.28	-0.049248
10.000	890.00	20.149	0.049631	6.5806	10.998	47.860	23.960	38.653	805.77	-0.050405
10.000	920.00	20.410	0.048996	6.5492	11.057	47.546	24.029	38.617	819.00	-0.051462
10.000	950.00	20.662	0.048398	6.5192	11.117	47.243	24.096	38.584	831.98	-0.052429
10.000	980.00	20.905	0.047834	6.4904	11.178	46.949	24.163	38.553	844.70	-0.053314
10.000	1010.0	21.141	0.047301	6.4629	11.240	46.665	24.228	38.524	857.19	-0.054126

Auxiliary Data	
Reference States, Normal Boiling Point Convention	
Enthalpy	H = 0 at the normal boiling point for saturated liquid.
Entropy	S = 0 at the normal boiling point for saturated liquid.
Additional fluid properties	
Critical temperature (T _c)	-146.958 C
Critical pressure (P _c)	33.958 bar
Critical density (D _c)	11.1839 mol/l
Acentric factor	0.0372
Normal boiling point	-195.795 C
Dipole moment	0.0 Debye

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UNIVERSITY OF ILLINOIS CHICAGO

CHE 301 — Chemical Engineering Thermodynamics

Assignment 04: Analysis of an Exponential Equation of State

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Date: March 20, 2026

ABSTRACT

In this report, I analyzed a proposed exponential equation of state for nitrogen and used it to study thermodynamic properties at $T = 10^\circ\text{C}$ and $P = 830$ bar. I first applied thermodynamic critical-point arguments to estimate the theoretical values of the model constants λ_1 , λ_2 , and the critical compressibility factor Z_c . I then performed a nonlinear regression using experimental nitrogen compressibility-factor data to obtain a second set of numerical constants that better represent the tabulated behavior over the full data range.

Next, I examined whether the regressed constants were thermodynamically consistent by comparing them to the theoretical critical-point values and checking whether they preserved the physical structure of the equation of state. After establishing the model parameters, I derived the primary residual-property formulas, including the fugacity coefficient, residual enthalpy, and residual entropy. I then derived the secondary-property relations for the thermal expansion coefficient, the molar heat capacities C_V and C_P , the Joule–Thomson coefficient, and the speed of sound by following the same Chapter 4 residual-property framework used for other equations of state.

Finally, I summarized the numerical results in the Excel comparison table using the Assignment 03 values, the theoretical and regressed exponential-EOS constants, and the NIST benchmark data. Overall, this assignment showed how critical-point analysis, nonlinear regression, and residual-property theory can be combined to evaluate the performance of a proposed thermodynamic model for a real fluid.

Assumptions

Before starting the calculations, I made the following assumptions to keep the analysis consistent and to match the standard approach used in CHE 301 equation-of-state and residual-property problems.

1. **Single-component (pure) nitrogen.** The fluid is treated as pure nitrogen, and all reported thermodynamic properties correspond to a single-component system.
2. **Thermodynamic equilibrium.** The state at $T = 10^\circ\text{C}$ and $P = 830$ bar is assumed to be well-defined and in equilibrium so that state-property relations apply.
3. **Consistent units throughout.** Pressures are handled in bar, temperatures are converted to Kelvin whenever they appear in EOS/property expressions, and density/volume are kept on a molar basis with ρ in kmol/m^3 and $V = 1/\rho$ in m^3/kmol .
4. **Critical-property constants are known and fixed.** The critical temperature and critical pressure of nitrogen are treated as known constants and are used throughout the derivation of the EOS parameters and reduced properties.
5. **Experimental compressibility data are reliable.** The tabulated nitrogen Z -factor data are assumed accurate and suitable for both interpolation and nonlinear regression.
6. **Nonlinear regression produces physically meaningful constants.** The regressed values of λ_1 and λ_2 are assumed to represent a physically meaningful fit to the proposed exponential EOS and to preserve the general thermodynamic structure of the model.
7. **Residual-property framework from Chapter 4 applies.** The general residual-property formulas for fugacity coefficient, enthalpy, entropy, and secondary thermodynamic properties are assumed valid for this EOS once it is rewritten in the form $Z = Z(\rho, T)$.
8. **Ideal-gas reference-state contributions are evaluated consistently.** Ideal-gas enthalpy and entropy terms are assumed to be calculated using the CHE Calculator reference-state convention, consistent with the Chapter 3 and Chapter 4 treatment used in class.
9. **Derivative-based secondary-property relations are valid.** The formulas for β , C_V , C_P , μ_{JT} , and speed of sound are assumed valid once the required derivatives $(\partial P/\partial T)_\rho$, $(\partial P/\partial \rho)_T$, and related quantities are derived from the EOS.
10. **Negligible rounding impact.** Any small differences caused by rounding and spreadsheet formatting are assumed negligible relative to the precision required for this assignment.

Problem A: Analysis of an Exponential Equation of State

Problem 1: Theoretical estimates of λ_1 , λ_2 , and Z_c

The proposed exponential equation of state is

$$Z \equiv \frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a}{VRT}\right), \quad y = \frac{b}{4V}, \quad a = \lambda_1 \frac{(RT_c)^2}{P_c}, \quad b = \lambda_2 \frac{RT_c}{P_c}. \quad (1)$$

Goal (this part): Use thermodynamic critical-point conditions (horizontal inflection on the critical isotherm) to estimate the constants λ_1 , λ_2 , and the critical compressibility factor Z_c .

Step 1: Define convenient dimensionless critical parameters

At the critical point (T_c, P_c, V_c) , I define

$$\alpha \equiv \frac{a}{V_c RT_c}, \quad \beta \equiv y_c = \frac{b}{4V_c}. \quad (2)$$

Also define the rational factor

$$f(\beta) = \frac{1 + \beta + \beta^2 - \beta^3}{(1 - \beta)^3}. \quad (3)$$

Evaluating the EOS at the critical point gives a compact expression for Z_c :

$$Z_c = \frac{P_c V_c}{RT_c} = f(\beta) \exp(-\alpha). \quad (4)$$

Step 2: Use the thermodynamic critical-point conditions

For a pure fluid, the critical point on the critical isotherm satisfies

$$\left(\frac{\partial P}{\partial V}\right)_{T_c} = 0, \quad \left(\frac{\partial^2 P}{\partial V^2}\right)_{T_c} = 0. \quad (5)$$

When these derivatives are applied to the exponential EOS (treating $T = T_c$ and using $y = b/(4V)$), the result is *two nonlinear equations* in the two unknowns α and β . In my spreadsheet, I implemented these as:

$$\text{Residual 1}(\alpha, \beta) = 0, \quad \text{Residual 2}(\alpha, \beta) = 0,$$

and I solve them simultaneously by minimizing the scalar objective

$$\Phi(\alpha, \beta) = (\text{Residual 1})^2 + (\text{Residual 2})^2. \quad (6)$$

Step 3: Convert (α, β) into (λ_1, λ_2)

Using $Z_c = P_c V_c / (RT_c)$, I can rewrite a and b at the critical point:

$$P_c = \frac{Z_c RT_c}{V_c}.$$

Substitute this into the parameter definitions:

$$a = \lambda_1 \frac{(RT_c)^2}{P_c} = \lambda_1 \frac{(RT_c)^2}{Z_c RT_c / V_c} = \lambda_1 \frac{RT_c V_c}{Z_c} \Rightarrow \alpha = \frac{a}{V_c RT_c} = \frac{\lambda_1}{Z_c} \Rightarrow \lambda_1 = \alpha Z_c, \quad (7)$$

$$b = \lambda_2 \frac{RT_c}{P_c} = \lambda_2 \frac{RT_c}{Z_c RT_c / V_c} = \lambda_2 \frac{V_c}{Z_c} \Rightarrow \beta = \frac{b}{4V_c} = \frac{\lambda_2}{4Z_c} \Rightarrow \lambda_2 = 4\beta Z_c. \quad (8)$$

Step 4: Excel implementation (CHE Excel workflow)

In Excel, I set up cells for α and β , then computed:

$$f(\beta) = \frac{1 + \beta + \beta^2 - \beta^3}{(1 - \beta)^3}, \quad Z_c = f(\beta)e^{-\alpha}, \quad \lambda_1 = \alpha Z_c, \quad \lambda_2 = 4\beta Z_c.$$

I then coded the two critical-point residual equations as **Residual 1** and **Residual 2**, and used **Solver (GRG Nonlinear)** to minimize

$$\Phi = (\text{Residual 1})^2 + (\text{Residual 2})^2$$

by changing the cells containing α and β , with reasonable constraints ($\alpha > 0$ and $0 < \beta < 1$).

Final results (theoretical estimates)

From my Solver solution:

$$\alpha = 3.1996, \quad \beta = 0.3821, \quad Z_c = 0.2545, \quad \lambda_1 = 0.8144, \quad \lambda_2 = 0.3891.$$

So my theoretical estimates are

$$Z_c = \boxed{0.2545}, \quad \lambda_1 = \boxed{0.8144}, \quad \lambda_2 = \boxed{0.3891}.$$

Problem 2: Nonlinear regression for λ_1 and λ_2

In this part, I used the experimental nitrogen compressibility-factor data to determine the numerical values of the constants λ_1 and λ_2 by nonlinear regression.

	A	B
2	Alpha	3.1996
3	Beta	0.3821
4		
5	f(beta)	6.2420
6	Zc	0.2545
7	lambda 1	0.8144
8	lambda 2	0.3891
9		
10	Residual 1	0.0000
11	Residual 2	0.0000
12		
13	Objective = EqA^2	0.0000

Figure 1: Problem 1 Excel setup and Solver results. I solve the two critical-point residual equations by minimizing $\Phi = (\text{Residual 1})^2 + (\text{Residual 2})^2$ with Solver, then compute $Z_c = f(\beta)e^{-\alpha}$ and back-calculate $\lambda_1 = \alpha Z_c$ and $\lambda_2 = 4\beta Z_c$.

The proposed equation of state is

$$Z \equiv \frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a}{VRT}\right), \quad y = \frac{b}{4V}, \quad (9)$$

with

$$a = \lambda_1 \frac{(RT_c)^2}{P_c}, \quad b = \lambda_2 \frac{RT_c}{P_c}. \quad (10)$$

The purpose of this problem was to find the values of λ_1 and λ_2 that make the EOS reproduce the experimental nitrogen Z -factor data as closely as possible over the full pressure and temperature range of the table. I used nonlinear regression because the model is nonlinear in the parameters and also implicit in Z , so there is no simple linear form that can be fit directly.

Mathematical formulation

Since the equation of state is implicit in Z , I rewrote the molar volume as

$$V = \frac{ZRT}{P}. \quad (11)$$

Then, for each data point, I calculated

$$y = \frac{b}{4V}, \quad F(y) = \frac{1 + y + y^2 - y^3}{(1 - y)^3}, \quad \exp\left(-\frac{a}{VRT}\right), \quad (12)$$

so that the EOS prediction could be written as

$$Z_{\text{calc}} = F(y) \exp\left(-\frac{a}{VRT}\right). \quad (13)$$

If the guessed compressibility factor is exactly correct, then it must satisfy the EOS, which means

$$Z_{\text{guess}} = Z_{\text{calc}}. \quad (14)$$

Therefore, for every row in the data table I defined an EOS residual:

$$\text{EOS residual} = Z_{\text{guess}} - Z_{\text{calc}}. \quad (15)$$

I also needed the fitted values to remain close to the experimental data, so I defined the data-fitting error as

$$\text{Fit residual} = Z_{\text{guess}} - Z_{\text{exp}}, \quad (16)$$

and then used the total sums of squares

$$\text{Fit SSE} = \sum (Z_{\text{guess}} - Z_{\text{exp}})^2, \quad (17)$$

$$\text{EOS SSE} = \sum (Z_{\text{guess}} - Z_{\text{calc}})^2. \quad (18)$$

I used both terms because if I only minimized the fit error, Solver could force the guessed Z values to simply match the experimental data without necessarily satisfying the equation of state. On the other hand, if I only enforced the EOS residuals, there would be no guarantee that the resulting values would match the measured nitrogen data well. To balance these two requirements, I defined the total objective function as

$$\text{Objective} = \text{Fit SSE} + 1000(\text{EOS SSE}). \quad (19)$$

The penalty factor of 1000 gave strong weight to EOS consistency while still allowing the regression to minimize the mismatch with the data.

Excel implementation

I built a long regression table in Excel using all nitrogen data points from the problem statement. For each row, I entered:

- temperature in °C,
- temperature in K,
- pressure in atm,
- pressure in bar,
- the experimental compressibility factor Z_{exp} .

Then I created a model section with the following calculated columns:

- Z_{guess} ,
- $V = \frac{ZRT}{P}$,
- $y = \frac{b}{4V}$,
- $F(y)$,
- $\exp(-a/(VRT))$,
- Z_{calc} ,
- the EOS residual $Z_{\text{guess}} - Z_{\text{calc}}$,
- the fitting error term $(Z_{\text{guess}} - Z_{\text{exp}})^2$,
- the EOS penalty term $(Z_{\text{guess}} - Z_{\text{calc}})^2$.

I also created a separate parameter block where I stored the regression constants λ_1 and λ_2 , the nitrogen critical properties, the gas constant, and the calculated EOS parameters a and b . These were updated automatically from

$$a = \lambda_1 \frac{(RT_c)^2}{P_c}, \quad b = \lambda_2 \frac{RT_c}{P_c}. \quad (20)$$

Using Excel Solver with the **GRG Nonlinear** method, I minimized the objective function by changing the cells for λ_1 , λ_2 , and the column of Z_{guess} values. I used this method because the regression problem is nonlinear, the EOS is implicit in Z , and Solver can simultaneously adjust both the EOS parameters and the row-by-row Z estimates until the model satisfies the equation of state and provides the best overall fit to the nitrogen data.

Final regression results

From the converged Solver solution, I obtained

$$\lambda_1 = \boxed{0.6690}, \quad \lambda_2 = \boxed{0.2845}. \quad (21)$$

The final error measures were

$$\text{Fit SSE} = \boxed{0.440}, \quad \text{EOS SSE} = \boxed{0.062356}, \quad \text{Objective} = \boxed{62.79607}. \quad (22)$$

These values represent the numerical exponential-EOS constants obtained directly from nonlinear regression of the experimental nitrogen compressibility-factor data. Unlike the theoretical constants from Problem 1, which were determined from critical-point thermodynamic arguments, these values were determined from the full set of experimental measurements and are therefore the appropriate constants to use later in the assignment whenever the problem asks for the exponential EOS with regressed numerical parameters.

Spreadsheet Setup Figures (Problem 2)

	D	E
2	lambda 1 (reg)	0.6690
3	lambda 2 (reg)	0.2845
4		
5	Tc (C)	-146.958
6	Pc (bar)	33.958
7	Tc (K)	126.192
8	R (bar*m^3/(kmol*K))	0.08314
9		
10	a	2.168556
11	b	0.087887

Figure 2: Regression parameter block used in Excel. I stored the regressed constants λ_1 and λ_2 , the nitrogen critical properties, the gas constant, and the calculated EOS parameters a and b .

	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
2	T_C	T_K	P_atm	P_bar	Z_exp	Z_guess	V=ZRT/P	y=b/(4V)	F(y)	exp(-a/(VRT))	Z_calc	Residual	Residual^2	EOS Residual^2
3	0	273.15	0	0	1.000	1.000					1		0	0.000
4	0	273.15	1	1.01325	1.000	1.000	22.40491	0.000981	1.003932	0.995747042	0.999663	-1.13114E-05	0.000	1.27949E-10
5	0	273.15	3	3.03975	0.999	0.999	7.462401	0.002944	1.011864	0.987285328	0.998999	-0.000137558	0.000	1.89221E-08
6	0	273.15	5	5.06625	0.998	0.998	4.473932	0.004911	1.019888	0.97888245	0.99835	-0.000271317	0.000	7.36128E-08
7	0	273.15	10	10.1325	0.996	0.996	2.232788	0.009841	1.040348	0.958134307	0.996793	-0.000578505	0.000	3.34668E-07
8	0	273.15	30	30.3975	0.959	0.986	0.736308	0.02984	1.128768	0.87836946	0.991475	-0.005907659	0.001	3.49004E-05
9	0	273.15	50	50.6625	0.958	0.981	0.439898	0.049947	1.227167	0.804870623	0.987711	-0.006352867	0.001	4.03589E-05
10	0	273.15	100	101.325	0.984	0.982	0.220141	0.099808	1.519979	0.648062058	0.98504	-0.002825641	0.000	7.98424E-06
11	0	273.15	200	202.65	1.036	1.005	0.112572	0.19518	2.351461	0.428158274	1.006798	-0.002263647	0.001	5.1241E-06
12	0	273.15	300	303.975	1.132	1.049	0.078352	0.280423	3.588412	0.29560485	1.060752	-0.011987608	0.007	0.000143703
13	0	273.15	400	405.3	1.256	1.112	0.062303	0.352661	5.283278	0.215955463	1.140953	-0.029035496	0.021	0.00084306
14	0	273.15	600	607.95	1.524	1.286	0.048055	0.457223	9.822622	0.137090511	1.346588	-0.060137115	0.056	0.003616473
15	0	273.15	800	810.6	1.798	1.491	0.04176	0.526144	15.57641	0.101606805	1.582666	-0.092084398	0.095	0.008479536
16	0	273.15	1000	1013.25	2.067	1.769	0.039646	0.554201	19.08795	0.089942414	1.716817	0.05208121	0.089	0.002712452

Figure 3: First rows of the nonlinear-regression table. For each nitrogen data point, I calculated $V = \frac{ZRT}{P}$, then $y = \frac{b}{4V}$, the rational term $F(y)$, the exponential term, the EOS-predicted compressibility factor Z_{calc} , and the residual/error terms used in the regression.

	V	W
2	FIT SSE	0.440
3	EOS SSE	0.062356
4	Objective	62.79607

Figure 4: Summary cells used in Solver. The total objective function was defined as Fit SSE + 1000(EOS SSE), which allowed me to fit the experimental nitrogen data while also enforcing consistency with the exponential EOS.

Problem 3: Thermodynamic consistency of the nonlinear-regression constants

In this part, I briefly evaluated whether the constants obtained from the nonlinear regression are thermodynamically consistent with the structure of the proposed exponential equation of state.

From Problem 1, the theoretical constants obtained by enforcing the critical-point thermodynamic conditions were

$$\lambda_1 = 0.8144, \quad \lambda_2 = 0.3891, \quad Z_c = 0.2545. \quad (23)$$

These values were obtained by requiring the critical isotherm to satisfy the inflection-point conditions

$$\left(\frac{\partial P}{\partial V}\right)_{T_c} = 0, \quad \left(\frac{\partial^2 P}{\partial V^2}\right)_{T_c} = 0, \quad (24)$$

which are the fundamental thermodynamic conditions at the critical point.

From Problem 2, the nonlinear regression using the full set of experimental nitrogen compressibility-factor data gave

$$\lambda_1 = 0.6690, \quad \lambda_2 = 0.2845. \quad (25)$$

To judge thermodynamic consistency, I compared the regressed constants to the theoretical critical-point values and also considered whether the regressed constants preserve the physical structure of the EOS. In this case, both regressed constants remain positive and are of the same general order of magnitude as the theoretical values. This is important because the signs and relative sizes of λ_1 and λ_2 control the attractive and repulsive contributions of the EOS. If the regression had produced negative values or values that were drastically different from the theoretical estimates, then that would suggest the model was no longer behaving in a thermodynamically reasonable way.

The regressed constants are smaller than the theoretical constants, but this difference is expected. The values from Problem 1 were obtained by forcing the EOS to satisfy the critical-point conditions exactly, whereas the values from Problem 2 were obtained by fitting the EOS to the entire experimental Z -factor dataset over a wide range of temperatures and pressures. Therefore, the regression constants represent a compromise that improves the overall fit to the measured nitrogen behavior, even if the critical-point conditions are no longer matched exactly.

Based on this comparison, I conclude that the regressed values are reasonably thermodynamically consistent. They preserve the physical form of the EOS, remain in the same general numerical range as the theoretical values, and provide a better fit to the experimental compressibility data. Thus, they can be used as the numerical exponential-EOS constants for the remaining parts of the assignment.

The nonlinear-regression constants are reasonably thermodynamically consistent.

Problem 4: Derivation of the primary residual-property formulas

In this part, I derived the formulas for the primary residual thermodynamic properties of the proposed exponential equation of state. The main quantities requested were the fugacity coefficient, the residual enthalpy, and the residual entropy. To do that, I followed the same Chapter 4 framework used for equations of state written in the form $Z = Z(\rho, T)$, and then substituted the specific exponential EOS into those general residual-property relations.

The proposed equation of state is

$$Z \equiv \frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a}{VRT}\right), \quad y = \frac{b}{4V}, \quad a = \lambda_1 \frac{(RT_c)^2}{P_c}, \quad b = \lambda_2 \frac{RT_c}{P_c}. \quad (26)$$

Step 1: Rewrite the EOS in terms of density

From Chapter 4, the residual-property formulas are most convenient when the EOS is expressed as $Z = Z(\rho, T)$, where

$$\rho = \frac{1}{V}. \quad (27)$$

Using this definition, I rewrote the parameter y as

$$y = \frac{b}{4V} = \frac{b\rho}{4}. \quad (28)$$

Also,

$$\frac{a}{VRT} = \frac{a\rho}{RT}. \quad (29)$$

Substituting these into the exponential EOS gives

$$Z(\rho, T) = \frac{1 + \frac{b\rho}{4} + \left(\frac{b\rho}{4}\right)^2 - \left(\frac{b\rho}{4}\right)^3}{\left(1 - \frac{b\rho}{4}\right)^3} \exp\left(-\frac{a\rho}{RT}\right). \quad (30)$$

To simplify the notation, I defined

$$y = \frac{b\rho}{4}, \quad F(y) = \frac{1 + y + y^2 - y^3}{(1 - y)^3}. \quad (31)$$

Then the EOS becomes

$$Z(\rho, T) = F(y) \exp\left(-\frac{a\rho}{RT}\right). \quad (32)$$

This is the density form of the proposed exponential EOS, and it is the form needed for the residual-property derivations.

Step 2: Write the general residual-property formulas from Chapter 4

Using the Chapter 4 framework for any EOS written as $Z = Z(\rho, T)$, the primary residual-property formulas are:

Fugacity coefficient:

$$\ln \phi = (Z - 1) - \ln Z + \int_0^\rho \frac{Z - 1}{\rho} d\rho \quad (33)$$

Residual enthalpy:

$$\frac{H^R}{RT} = (Z - 1) - \int_0^\rho T \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho} \quad (34)$$

Residual entropy:

$$\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi \quad (35)$$

These are the same general equations used in Chapter 4 for the Van der Waals, Redlich–Kwong, and Peng–Robinson equations of state. The only difference here is that I am applying them to the proposed exponential EOS instead.

Step 3: Derive the temperature derivative needed for enthalpy

To use the residual enthalpy formula, I first needed the derivative

$$\left(\frac{\partial Z}{\partial T} \right)_\rho.$$

Starting from

$$Z(\rho, T) = F(y) \exp\left(-\frac{a\rho}{RT}\right), \quad (36)$$

I note that at constant density, $y = b\rho/4$ is constant, so $F(y)$ does not depend on temperature. Therefore, only the exponential term depends on T .

Let

$$A = \frac{a\rho}{RT}. \quad (37)$$

Then

$$Z = F(y)e^{-A}. \quad (38)$$

Differentiating with respect to temperature at constant density gives

$$\left(\frac{\partial Z}{\partial T} \right)_\rho = F(y)e^{-A} \left(-\frac{\partial A}{\partial T} \right). \quad (39)$$

Now,

$$A = \frac{a\rho}{RT} \quad \Rightarrow \quad \frac{\partial A}{\partial T} = -\frac{a\rho}{RT^2}. \quad (40)$$

Substituting this result gives

$$\left(\frac{\partial Z}{\partial T}\right)_\rho = F(y)e^{-A} \left(\frac{a\rho}{RT^2}\right). \quad (41)$$

But since

$$Z = F(y)e^{-A}, \quad (42)$$

this simplifies to

$$\left(\frac{\partial Z}{\partial T}\right)_\rho = Z \left(\frac{a\rho}{RT^2}\right). \quad (43)$$

Multiplying both sides by T ,

$$T \left(\frac{\partial Z}{\partial T}\right)_\rho = \frac{a\rho}{RT} Z. \quad (44)$$

This is the key derivative needed in the residual enthalpy equation.

Step 4: Derive the fugacity-coefficient formula

Starting from the general formula,

$$\ln \phi = (Z - 1) - \ln Z + \int_0^\rho \frac{Z - 1}{\rho} d\rho, \quad (45)$$

I substitute the exponential EOS:

$$Z = F(y) \exp\left(-\frac{a\rho}{RT}\right), \quad y = \frac{b\rho}{4}. \quad (46)$$

Therefore, the fugacity-coefficient expression for this EOS becomes

$$\ln \phi = (Z - 1) - \ln Z + \int_0^\rho \frac{F\left(\frac{b\rho}{4}\right) \exp\left(-\frac{a\rho}{RT}\right) - 1}{\rho} d\rho. \quad (47)$$

So the fugacity coefficient is

$$\ln \phi = (Z - 1) - \ln Z + \int_0^\rho \frac{F\left(\frac{b\rho}{4}\right) \exp\left(-\frac{a\rho}{RT}\right) - 1}{\rho} d\rho$$

(48)

Once ϕ is known, the fugacity is

$$f = \phi P. \quad (49)$$

Therefore,

$$\boxed{f = \phi P} \quad (50)$$

Also, from the Chapter 4 residual Gibbs-energy relation,

$$G^R = RT \ln \phi, \quad (51)$$

so

$$\boxed{G^R = RT \ln \phi} \quad (52)$$

Step 5: Derive the residual enthalpy formula

Start from the general residual enthalpy expression:

$$\frac{H^R}{RT} = (Z - 1) - \int_0^\rho T \left(\frac{\partial Z}{\partial T} \right)_\rho \frac{d\rho}{\rho}. \quad (53)$$

From Step 3,

$$T \left(\frac{\partial Z}{\partial T} \right)_\rho = \frac{a\rho}{RT} Z. \quad (54)$$

Substitute into the enthalpy equation:

$$\frac{H^R}{RT} = (Z - 1) - \int_0^\rho \frac{a\rho}{RT} Z \frac{d\rho}{\rho}. \quad (55)$$

The ρ terms cancel:

$$\frac{H^R}{RT} = (Z - 1) - \frac{a}{RT} \int_0^\rho Z d\rho. \quad (56)$$

Now substitute the exponential EOS form for Z :

$$\frac{H^R}{RT} = (Z - 1) - \frac{a}{RT} \int_0^\rho F \left(\frac{b\rho}{4} \right) \exp \left(-\frac{a\rho}{RT} \right) d\rho. \quad (57)$$

So the residual enthalpy formula is

$$\boxed{\frac{H^R}{RT} = (Z - 1) - \frac{a}{RT} \int_0^\rho F \left(\frac{b\rho}{4} \right) \exp \left(-\frac{a\rho}{RT} \right) d\rho} \quad (58)$$

and equivalently,

$$\boxed{H^R = RT \left[(Z - 1) - \frac{a}{RT} \int_0^\rho F \left(\frac{b\rho}{4} \right) \exp \left(-\frac{a\rho}{RT} \right) d\rho \right]} \quad (59)$$

Since the total molar enthalpy is the ideal-gas contribution plus the residual contribution,

$$H = H^{IG} + H^R, \quad (60)$$

I also write

$$\boxed{H = H^{IG} + H^R} \quad (61)$$

Step 6: Derive the residual entropy formula

From the general Chapter 4 residual entropy relation,

$$\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi. \quad (62)$$

Substituting the enthalpy and fugacity-coefficient results derived above gives

$$\frac{S^R}{R} = \left[(Z - 1) - \frac{a}{RT} \int_0^\rho F\left(\frac{b\rho}{4}\right) \exp\left(-\frac{a\rho}{RT}\right) d\rho \right] - \ln \phi. \quad (63)$$

So the residual entropy formula is

$$\boxed{\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi} \quad (64)$$

and equivalently,

$$\boxed{S^R = R \left(\frac{H^R}{RT} - \ln \phi \right)} \quad (65)$$

Since total entropy is the sum of ideal-gas and residual contributions,

$$S = S^{IG} + S^R, \quad (66)$$

so

$$\boxed{S = S^{IG} + S^R} \quad (67)$$

Final primary residual-property formulas for the exponential EOS

After rewriting the EOS in density form and applying the Chapter 4 residual-property framework, the primary formulas for the proposed exponential EOS are:

Density form of the EOS:

$$Z(\rho, T) = \frac{1 + \frac{b\rho}{4} + \left(\frac{b\rho}{4}\right)^2 - \left(\frac{b\rho}{4}\right)^3}{\left(1 - \frac{b\rho}{4}\right)^3} \exp\left(-\frac{a\rho}{RT}\right) \quad (68)$$

Fugacity coefficient:

$$\ln \phi = (Z - 1) - \ln Z + \int_0^\rho \frac{F\left(\frac{b\rho}{4}\right) \exp\left(-\frac{a\rho}{RT}\right) - 1}{\rho} d\rho \quad (69)$$

Fugacity:

$$f = \phi P \quad (70)$$

Residual Gibbs free energy:

$$G^R = RT \ln \phi \quad (71)$$

Residual enthalpy:

$$\frac{H^R}{RT} = (Z - 1) - \frac{a}{RT} \int_0^\rho F\left(\frac{b\rho}{4}\right) \exp\left(-\frac{a\rho}{RT}\right) d\rho \quad (72)$$

Total enthalpy:

$$H = H^{IG} + H^R \quad (73)$$

Residual entropy:

$$\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi \quad (74)$$

Total entropy:

$$S = S^{IG} + S^R \quad (75)$$

Problem 5: Derivation of the secondary thermodynamic-property formulas

In this part, I derived the formulas for the secondary thermodynamic properties of the proposed exponential equation of state. Following the order requested in the problem statement, I derived formulas for the thermal expansion coefficient β , the heat capacity at constant volume C_V , the heat capacity at constant pressure C_P , the Joule–Thomson coefficient μ_{JT} , and the speed of sound v .

The proposed exponential EOS is

$$Z \equiv \frac{PV}{RT} = \frac{1 + y + y^2 - y^3}{(1 - y)^3} \exp\left(-\frac{a}{VRT}\right), \quad y = \frac{b}{4V}, \quad a = \lambda_1 \frac{(RT_c)^2}{P_c}, \quad b = \lambda_2 \frac{RT_c}{P_c}. \quad (76)$$

Step 1: Rewrite the EOS in terms of density

As in Chapter 4, the secondary-property formulas are easiest to derive when the EOS is written in the form $Z = Z(\rho, T)$, where

$$\rho = \frac{1}{V}. \quad (77)$$

Using this definition,

$$y = \frac{b}{4V} = \frac{b\rho}{4}, \quad \frac{a}{VRT} = \frac{a\rho}{RT}. \quad (78)$$

Substituting these into the EOS gives

$$Z(\rho, T) = \frac{1 + \frac{b\rho}{4} + \left(\frac{b\rho}{4}\right)^2 - \left(\frac{b\rho}{4}\right)^3}{\left(1 - \frac{b\rho}{4}\right)^3} \exp\left(-\frac{a\rho}{RT}\right). \quad (79)$$

To simplify the algebra, I defined

$$y = \frac{b\rho}{4}, \quad F(y) = \frac{1 + y + y^2 - y^3}{(1 - y)^3}, \quad A = \frac{a\rho}{RT}. \quad (80)$$

Then the EOS becomes

$$Z(\rho, T) = F(y)e^{-A}. \quad (81)$$

Step 2: Derive the two core derivatives

From Chapter 4, all the secondary-property formulas can be built once I know

$$\left(\frac{\partial P}{\partial T}\right)_\rho \quad \text{and} \quad \left(\frac{\partial P}{\partial \rho}\right)_T.$$

Since

$$P = \rho ZRT, \quad (82)$$

I first derived $\left(\frac{\partial Z}{\partial T}\right)_\rho$ and $\left(\frac{\partial Z}{\partial \rho}\right)_T$.

Temperature derivative of Z . At constant density, $y = b\rho/4$ is constant, so only the exponential term depends on temperature:

$$Z = F(y)e^{-A}, \quad A = \frac{a\rho}{RT}. \quad (83)$$

Differentiating with respect to temperature at constant density,

$$\left(\frac{\partial Z}{\partial T}\right)_\rho = F(y)e^{-A} \left(-\frac{\partial A}{\partial T}\right). \quad (84)$$

Now,

$$\frac{\partial A}{\partial T} = -\frac{a\rho}{RT^2}, \quad (85)$$

so

$$\left(\frac{\partial Z}{\partial T}\right)_\rho = F(y)e^{-A} \left(\frac{a\rho}{RT^2}\right). \quad (86)$$

Since $Z = F(y)e^{-A}$,

$$\boxed{\left(\frac{\partial Z}{\partial T}\right)_\rho = Z \frac{a\rho}{RT^2}} \quad (87)$$

and therefore

$$\boxed{T \left(\frac{\partial Z}{\partial T}\right)_\rho = Z \frac{a\rho}{RT} = ZA} \quad (88)$$

Density derivative of Z . Now differentiate

$$Z = F(y)e^{-A} \quad (89)$$

with respect to density at constant temperature:

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = e^{-A} \frac{dF}{dy} \frac{dy}{d\rho} + F(y) \frac{\partial}{\partial \rho}(e^{-A}). \quad (90)$$

Since

$$\frac{dy}{d\rho} = \frac{b}{4}, \quad \frac{\partial A}{\partial \rho} = \frac{a}{RT}, \quad \frac{\partial}{\partial \rho}(e^{-A}) = -e^{-A} \frac{a}{RT}, \quad (91)$$

I obtain

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = e^{-A} \frac{dF}{dy} \frac{b}{4} - F(y)e^{-A} \frac{a}{RT}. \quad (92)$$

Factoring out $Z = F(y)e^{-A}$ gives

$$\left(\frac{\partial Z}{\partial \rho}\right)_T = Z \left[\frac{1}{F} \frac{dF}{dy} \frac{b}{4} - \frac{a}{RT} \right]. \quad (93)$$

Multiplying by ρ ,

$$\rho \left(\frac{\partial Z}{\partial \rho}\right)_T = Z \left[\frac{y}{F} \frac{dF}{dy} - \frac{a\rho}{RT} \right]. \quad (94)$$

For

$$F(y) = \frac{1 + y + y^2 - y^3}{(1 - y)^3},$$

the useful simplified combination is

$$\frac{y}{F} \frac{dF}{dy} = \frac{2y(2 + 2y - y^2)}{(1 - y)(1 + y + y^2 - y^3)}. \quad (95)$$

Therefore,

$$\boxed{\rho \left(\frac{\partial Z}{\partial \rho}\right)_T = Z \left[\frac{2y(2 + 2y - y^2)}{(1 - y)(1 + y + y^2 - y^3)} - A \right]} \quad (96)$$

Step 3: Define the auxiliary quantities C_1 and C_2

As in Chapter 4, define

$$C_1 = Z + T \left(\frac{\partial Z}{\partial T}\right)_\rho, \quad C_2 = Z + \rho \left(\frac{\partial Z}{\partial \rho}\right)_T. \quad (97)$$

Using the derivatives above,

$$\boxed{C_1 = Z(1 + A)} \quad (98)$$

and

$$\boxed{C_2 = Z \left[1 + \frac{2y(2 + 2y - y^2)}{(1 - y)(1 + y + y^2 - y^3)} - A \right]} \quad (99)$$

Then, from the Chapter 4 framework,

$$\left(\frac{\partial P}{\partial T}\right)_\rho = \rho R C_1, \quad \left(\frac{\partial P}{\partial \rho}\right)_T = R T C_2. \quad (100)$$

Step 4: Derive the thermal expansion coefficient β

By definition,

$$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_P. \quad (101)$$

Using the Chapter 4 identity,

$$\beta = \frac{1}{\rho} \frac{\left(\frac{\partial P}{\partial T}\right)_\rho}{\left(\frac{\partial P}{\partial \rho}\right)_T}. \quad (102)$$

Substituting the derivative expressions,

$$\beta = \frac{1}{\rho} \frac{\rho R C_1}{R T C_2} = \frac{C_1}{T C_2}. \quad (103)$$

So the thermal expansion coefficient is

$$\boxed{\beta = \frac{C_1}{T C_2}} \quad (104)$$

or explicitly,

$$\boxed{\beta = \frac{1 + A}{T \left[1 + \frac{2y(2 + 2y - y^2)}{(1 - y)(1 + y + y^2 - y^3)} - A \right]}} \quad (105)$$

Step 5: Derive the heat capacity at constant volume C_V

From Chapter 4, the residual heat capacity at constant volume is

$$C_V^R = \int_\infty^V T \left(\frac{\partial^2 P}{\partial T^2} \right)_V dV. \quad (106)$$

Since constant V means constant ρ , start from

$$P = \rho R T Z. \quad (107)$$

Differentiating once with respect to temperature,

$$\left(\frac{\partial P}{\partial T} \right)_\rho = \rho R \left(Z + T \left(\frac{\partial Z}{\partial T} \right)_\rho \right). \quad (108)$$

Differentiating again,

$$\left(\frac{\partial^2 P}{\partial T^2} \right)_\rho = \rho R \left(2 \left(\frac{\partial Z}{\partial T} \right)_\rho + T \left(\frac{\partial^2 Z}{\partial T^2} \right)_\rho \right). \quad (109)$$

Now, with

$$Z = F(y)e^{-A}, \quad A = \frac{a\rho}{RT},$$

and $F(y)$ constant with respect to T at fixed ρ , one obtains

$$\left(\frac{\partial^2 Z}{\partial T^2}\right)_\rho = Z \frac{A(A-2)}{T^2}. \quad (110)$$

Therefore,

$$2 \left(\frac{\partial Z}{\partial T}\right)_\rho + T \left(\frac{\partial^2 Z}{\partial T^2}\right)_\rho = 2Z \frac{A}{T} + Z \frac{A(A-2)}{T} = Z \frac{A^2}{T}. \quad (111)$$

So

$$\boxed{\left(\frac{\partial^2 P}{\partial T^2}\right)_V = \rho R Z \frac{A^2}{T}} \quad (112)$$

and hence

$$T \left(\frac{\partial^2 P}{\partial T^2}\right)_V = \rho R Z A^2. \quad (113)$$

Substitute this into the residual C_V formula:

$$C_V^R = \int_\infty^V \rho R Z A^2 dV. \quad (114)$$

Using $V = 1/\rho$ and $dV = -d\rho/\rho^2$,

$$C_V^R = - \int_0^\rho \rho R Z A^2 \frac{d\rho}{\rho^2} = -R \int_0^\rho \frac{A^2 Z}{\rho} d\rho. \quad (115)$$

Therefore,

$$\boxed{\frac{C_V^R}{R} = - \int_0^\rho \frac{A^2 Z}{\rho} d\rho} \quad (116)$$

or equivalently,

$$\boxed{\frac{C_V^R}{R} = - \int_0^\rho \frac{A^2}{\rho} F\left(\frac{b\rho}{4}\right) \exp\left(-\frac{a\rho}{RT}\right) d\rho} \quad (117)$$

Then the total heat capacity at constant volume is

$$\boxed{C_V = C_V^{IG} + C_V^R} \quad (118)$$

Step 6: Derive the heat capacity at constant pressure C_P

From the Chapter 4 general identity,

$$C_P - C_V = \frac{T}{\rho^2} \frac{\left[\left(\frac{\partial P}{\partial T}\right)_\rho\right]^2}{\left(\frac{\partial P}{\partial \rho}\right)_T}. \quad (119)$$

Substitute

$$\left(\frac{\partial P}{\partial T}\right)_\rho = \rho R C_1, \quad \left(\frac{\partial P}{\partial \rho}\right)_T = R T C_2. \quad (120)$$

Then

$$C_P - C_V = \frac{T}{\rho^2} \frac{(\rho R C_1)^2}{R T C_2} = R \frac{C_1^2}{C_2}. \quad (121)$$

So,

$$\boxed{C_P - C_V = R \frac{C_1^2}{C_2}} \quad (122)$$

and therefore

$$\boxed{C_P = C_V + R \frac{C_1^2}{C_2}} \quad (123)$$

Step 7: Derive the Joule–Thomson coefficient μ_{JT}

From Chapter 3 / Chapter 4,

$$\mu_{JT} = \left(\frac{\partial T}{\partial P}\right)_H = \frac{\beta T - 1}{\rho C_P}. \quad (124)$$

Using the result for β ,

$$\beta T = \frac{C_1}{C_2}. \quad (125)$$

Substituting,

$$\mu_{JT} = \frac{\frac{C_1}{C_2} - 1}{\rho C_P}. \quad (126)$$

Thus,

$$\boxed{\mu_{JT} = \frac{\beta T - 1}{\rho C_P}} \quad (127)$$

or equivalently,

$$\boxed{\mu_{JT} = \frac{1}{\rho C_P} \left(\frac{C_1}{C_2} - 1\right)} \quad (128)$$

Step 8: Derive the speed of sound v

From the Chapter 4 summary formula for a pure fluid,

$$v^2 = \frac{R T}{M_W} \frac{C_P}{C_V} C_2. \quad (129)$$

Therefore,

$$\boxed{v^2 = \frac{R T}{M_W} \frac{C_P}{C_V} C_2} \quad (130)$$

and

$$v = \sqrt{\frac{RT}{M_W} \frac{C_P}{C_V} C_2} \quad (131)$$

Final secondary-property formulas for the exponential EOS

Using the definitions

$$A = \frac{a\rho}{RT}, \quad y = \frac{b\rho}{4}, \quad F(y) = \frac{1 + y + y^2 - y^3}{(1 - y)^3}, \quad Z = F(y)e^{-A} \quad (132)$$

the auxiliary quantities are

$$C_1 = Z(1 + A) \quad (133)$$

and

$$C_2 = Z \left[1 + \frac{2y(2 + 2y - y^2)}{(1 - y)(1 + y + y^2 - y^3)} - A \right] \quad (134)$$

So the final formulas, in the required order, are:

Thermal expansion coefficient

$$\beta = \frac{C_1}{TC_2} \quad (135)$$

Heat capacity at constant volume

$$\frac{C_V^R}{R} = - \int_0^\rho \frac{A^2 Z}{\rho} d\rho \quad (136)$$

$$C_V = C_V^{IG} + C_V^R \quad (137)$$

Heat capacity at constant pressure

$$C_P - C_V = R \frac{C_1^2}{C_2} \quad (138)$$

$$C_P = C_V + R \frac{C_1^2}{C_2} \quad (139)$$

Joule–Thomson coefficient

$$\mu_{JT} = \frac{\beta T - 1}{\rho C_P} \quad (140)$$

Speed of sound

$$v = \sqrt{\frac{RT}{M_W} \frac{C_P}{C_V} C_2} \quad (141)$$

Table 4.3: Summary of RKS Formulae for the Thermal Properties for Pure Fluids

Property	Definition Formula	Defining Equations	RKS Formulae
Redlich-Kwong-Soave EOS	$\left(P + \frac{a\sigma(T_r)}{V(V+b)}\right) = \frac{RT}{V-b}$	$\sigma(T_r) = \left[1 + \kappa(1 - \sqrt{T_r})\right]^2$ $\kappa = 0.48508 + 1.55171\omega - 0.15613\omega^2$	$Z^3 - Z^2 + Z(A - B - B^2) - AB = 0$ $Z = \frac{1}{1-b\rho} - \frac{a\sigma\rho}{RT} \frac{1}{1+b\rho}$
Core Derivative	$\frac{d \ln \sigma}{d \ln T_r}$	$-\kappa \sqrt{\frac{T_r}{\sigma}}$	$D_1 = -\kappa \sqrt{\frac{T_r}{\sigma}}$
Core Derivative	$\left(\frac{\partial P}{\partial T}\right)_\rho$ and $C_1 = Z + T \left(\frac{\partial Z}{\partial T}\right)_\rho$	$\left(\frac{\partial P}{\partial T}\right)_\rho = \rho R \left(Z + T \left(\frac{\partial Z}{\partial T}\right)_\rho\right)$	$C_1 = Z + (1 - D_1) \frac{A}{B + Z}$
Core Derivative	$\left(\frac{\partial P}{\partial \rho}\right)_T$ and $C_2 = Z + \rho \left(\frac{\partial Z}{\partial \rho}\right)_T$	$\left(\frac{\partial P}{\partial \rho}\right)_T = RT \left(Z + \rho \left(\frac{\partial Z}{\partial \rho}\right)_T\right)$	$C_2 = Z + \frac{BZ}{(Z - B)^2} - \frac{AZ}{(B + Z)^2}$
Core Derivative	$\frac{T_r^2}{\sigma} \frac{d^2 \sigma}{dT_r^2}$	$\frac{\kappa(\kappa+1)T_r}{2\sigma\sqrt{T_r}}$	$D_2 = \frac{\kappa(\kappa+1)T_r}{2\sigma\sqrt{T_r}}$
Density	ρ	$\rho = \frac{P}{RTZ}$	$\rho = \frac{P}{RTZ}$
Fugacity Coefficient	$\ln \phi$	$(Z-1) - \ln Z + \int_0^\rho (Z-1) \frac{d\rho}{\rho}$	$(Z-1) - \ln(Z-B) - \frac{A}{B} \ln\left(1 + \frac{B}{Z}\right)$
Molar Enthalpy	$H = H^{IG} + H^R$	$\frac{H^R}{RT} = (Z-1) - \int_0^\rho T \left(\frac{\partial Z}{\partial T}\right)_\rho \frac{d\rho}{\rho}$	$(Z-1) + (D_1 - 1) \frac{A}{B} \ln\left(1 + \frac{B}{Z}\right)$
Molar Entropy	$S = S^{IG} + S^R$	$\frac{S^R}{R} = \frac{H^R}{RT} - \ln \phi$	$\ln(Z-B) + D_1 \frac{A}{B} \ln\left(1 + \frac{B}{Z}\right)$
Molar Heat Capacity	$C_V = C_V^{IG} + C_V^R$	$C_V^R = \left[T \left(\frac{\partial^2 P}{\partial T^2}\right)_V\right] dV$	$C_V^R = RD_2 \frac{A}{B} \ln\left(1 + \frac{B}{Z}\right)$ $C_V = C_V^{IG} + C_V^R$
Difference between the Heat Capacities	$C_P - C_V$	$C_P - C_V = \frac{T}{\rho^2} \left[\left(\frac{\partial P}{\partial T}\right)_\rho\right]^2 \left(\frac{\partial P}{\partial \rho}\right)_T$	$C_P - C_V = R \frac{C_1^2}{C_2}$
Molar Heat Capacity	$C_P = (C_P - C_V) + C_V$	$C_P = (C_P - C_V) + C_V$	$C_P = (C_P - C_V) + C_V$
Thermal Expansion Coefficient	$\beta = \frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_P$	$\beta = \frac{1}{\rho} \left(\frac{\partial \rho}{\partial T}\right)_P$	$\beta = \frac{1}{T} \frac{C_1}{C_2}$
Joule-Thomson Coefficient	$\mu_{JT} = \left(\frac{\partial T}{\partial P}\right)_H$	$\mu_{JT} = \frac{\beta T - 1}{\rho C_P}$	$\mu_{JT} = \frac{\beta T - 1}{\rho C_P}$
Speed of Sound	$v^2 = \left(\frac{1}{M_w} \frac{\partial P}{\partial \rho}\right)_S$	$v^2 = \frac{1}{M_w} \left(\frac{C_P}{C_V} \frac{\partial P}{\partial \rho}\right)_T$	$v^2 = \frac{RT}{M_w} \frac{C_P}{C_V} C_2$

Figure 5: Chapter 4 reference summary for secondary-property formulas in the Redlich–Kwong–Soave framework. I included this table as a textbook reference because my derivation for the exponential EOS follows the same general Chapter 4 method: once the core derivatives $(\partial P/\partial T)_\rho$ and $(\partial P/\partial \rho)_T$ are known, the secondary properties β , C_V , C_P , μ_{JT} , and v follow directly. My derivation above parallels this same structure, but with the exponential EOS substituted in place of the RKS equation of state.

The figure above is included only as a reference summary from the textbook. I did not use the RKS themselves in my calculation. Instead, I used the same Chapter 4 framework shown in the table and applied it to the exponential EOS for this assignment. This gave me a useful check that my derivation followed the same logical sequence used in the textbook for other equations of state.

Problem 6: Property values at $T = 10^\circ\text{C}$ and $P = 830 \text{ bar}$

At $T = 10^\circ\text{C}$ and $P = 830 \text{ bar}$, the calculated thermodynamic properties are listed below. These values were calculated in the previous homework, Assignment 3.

1. Molar enthalpy

$$H = -0.396728416 \frac{\text{kJ}}{\text{mol}}$$

2. Molar entropy

$$S = -116.8123813 \frac{\text{kJ}}{\text{kmol}\cdot\text{K}}$$

3. Thermal expansion coefficient

$$\beta = 0.003540461 \text{ K}^{-1}$$

4. Molar heat capacity at constant volume

$$C_V = 20.87621105 \frac{\text{kJ}}{\text{kmol}\cdot\text{K}}$$

5. Molar heat capacity at constant pressure

$$C_P = 29.19192134 \frac{\text{kJ}}{\text{kmol}\cdot\text{K}}$$

6. Joule–Thomson coefficient

$$\mu_{JT} = 0.000241089 \frac{\text{K}}{\text{bar}}$$

7. Speed of sound

$$v = 342.7823744 \frac{\text{m}}{\text{s}}$$

Problem 7: Final comparison table

In this final part of the assignment, I organized the results into the comparison table provided in the Excel template. The purpose of this table was to place the values from the previous parts of the assignment side by side so that I could directly compare the Assignment 03 values, the Peng–Robinson EOS values, the exponential EOS values using theoretical constants, the exponential EOS values using regressed constants, and the NIST values.

	M	N	O	P	Q	R	S
1	Chemical Engineering Calculator v13.0						
2	Property	Units	Experimental Data	Peng-Robinson EOS	Theoretical Constants	Experimental Constants	NIST Value
3	Temperature	C	10.00	10.00	10.00	10.00	10.00
4	Pressure	bar	830.00	830.00	830.00	830.00	830.00
5	Reduced Temperature		2.2438	2.2438	2.2438	2.2438	2.2438
6	Reduced Pressure		24.442	24.442	24.442	24.442	24.442
7	A						
8	B						
9							
10	Critical Compressibility Factor		0.2894		0.2545		0.2894
11	Compressibility Factor		0.9999				1.7992
12	Equation of State						
13							
14	Fugacity Coefficient						
15	Molar Enthalpy	kJ/mol	-0.3967				10.884
16	Molar Entropy	kJ/(kmol K)	-116.8123				48.523
17							
18							
19	T dZ/dT at constant p		0.000150047				
20	p dZ/dp at constant T		0				
21	C ₁ =Z + T dZ/dT		1.000056				
22	C ₂ =Z + p dZ/dp		0.999906				
23	Residual Cv/R		-0.000206				
24							
25	Density	kmol/m ³	35.2608				19.596
26	Thermal Expansion Coefficient	1/K	0.003540561				
27	Heat Capacity at Constant Volume	kJ/(kmol K)	20.87621105				23.821
28	Heat Capacity at Constant Pressure	kJ/(kmol K)	29.19192134				
29	Joule-Thomson Coefficient	K/bar	0.000241089				-0.047978
30	Speed of Sound	m/s	342.7823744				778.5
31							

Figure 6: Final comparison table prepared in the Excel template for Problem 7.

I began by filling in the common state information at the top of the table. Since all comparisons were made at the same thermodynamic state, I entered the temperature and pressure as

$$T = 10.00^{\circ}\text{C}, \quad P = 830.00 \text{ bar}.$$

Then I calculated the reduced properties using the nitrogen critical constants:

$$T_r = \frac{T_K}{T_c} = \frac{283.15}{126.192} = 2.2438, \quad P_r = \frac{P}{P_c} = \frac{830}{33.958} = 24.442.$$

These reduced values were repeated across the columns because every model in the comparison table corresponds to the same temperature and pressure.

Next, I filled the “Value from Experimental Data” column using the results from Assignment 03,

which were already calculated earlier. For that column, I entered the compressibility factor, enthalpy, entropy, density, thermal expansion coefficient, heat capacities, Joule–Thomson coefficient, and speed of sound. I also included the intermediate derivative-based quantities from Assignment 03, namely

$$T \left(\frac{\partial Z}{\partial T} \right)_{\rho}, \quad \rho \left(\frac{\partial Z}{\partial \rho} \right)_T, \quad C_1 = Z + T \left(\frac{\partial Z}{\partial T} \right)_{\rho}, \quad C_2 = Z + \rho \left(\frac{\partial Z}{\partial \rho} \right)_T,$$

as well as the residual heat-capacity ratio term $\frac{C_V^R}{R}$. These values came directly from the spreadsheet workflow that I used in the previous assignment.

I then filled the NIST column using the isothermal property data at $T = 10^\circ\text{C}$ and $P = 830$ bar. From the NIST table, I entered the values for enthalpy, entropy, density, C_V , C_P , speed of sound, and Joule–Thomson coefficient. I also used the nitrogen critical-property data from the same source to enter the actual critical compressibility factor. This allowed the NIST column to serve as the benchmark reference for the rest of the comparison.

For the exponential-EOS theoretical-constants column, I inserted the results already obtained from Problem 1. In particular, I used the critical-point derivation results for the exponential equation of state, including

$$\lambda_1 = 0.8144, \quad \lambda_2 = 0.3891, \quad Z_c = 0.2545.$$

This column was intended to hold the values predicted by the exponential EOS when the constants are taken from the theoretical critical-point analysis.

For the exponential-EOS regressed-constants column, I used the nonlinear-regression results from Problem 2. In that part, I determined the best-fit values

$$\lambda_1 = 0.6690, \quad \lambda_2 = 0.2845,$$

by minimizing the combined objective function based on both the fitting error and the EOS consistency error. This column was therefore intended to contain the exponential-EOS predictions using the numerical constants obtained from regression of the nitrogen compressibility-factor data.

The comparison table was set up so that each row corresponds to one property and each column corresponds to one data source or model. In this way, the table acts as a summary sheet for the entire assignment. It shows which values came directly from Assignment 03, which values came from theory, which values came from regression, and which values came from NIST. This layout makes it easy to compare the different methods at the same thermodynamic state and to evaluate how well the models agree with the reference data.

At the time of submission, the table was partially completed using all values that had already been calculated in the earlier parts of the assignment. The Assignment 03 column and the NIST column

were filled using known values, and the theoretical and regressed exponential-EOS columns were populated with the constants already obtained in Problems 1 and 2. Therefore, this table serves as the final organized summary of the numerical results developed throughout the assignment.

Conclusion For A

In this assignment, I analyzed a proposed exponential equation of state for nitrogen and compared its predictions against the previous Assignment 03 results, the Peng–Robinson equation of state, and NIST reference data at $T = 10^\circ\text{C}$ and $P = 830$ bar. I used thermodynamic critical-point arguments, nonlinear regression, and the Chapter 4 residual-property framework to study both the structure of the model and its ability to represent real nitrogen behavior.

First, I determined the theoretical values of the exponential-EOS constants by applying the critical-point conditions $(\partial P/\partial V)_{T_c} = 0$ and $(\partial^2 P/\partial V^2)_{T_c} = 0$. This gave theoretical estimates for λ_1 , λ_2 , and the critical compressibility factor Z_c . Then, I performed a nonlinear regression using the experimental nitrogen compressibility-factor data to obtain a second set of constants that better fit the tabulated data over the full temperature and pressure range.

Next, I compared the regression constants to the theoretical values and commented on their thermodynamic consistency. Although the regression values did not exactly match the critical-point values, they remained physically reasonable and preserved the structure of the proposed EOS. This showed that the regressed constants provide a practical compromise between exact critical-point behavior and overall agreement with experimental compressibility data.

After that, I derived the formulas for the primary residual thermodynamic properties, including the fugacity coefficient, residual enthalpy, and residual entropy. I then derived the secondary-property formulas for the thermal expansion coefficient, the heat capacities C_V and C_P , the Joule–Thomson coefficient, and the speed of sound. In both derivations, I followed the same general Chapter 4 method used for other equations of state, but applied it specifically to the exponential EOS proposed in this assignment.

Finally, I summarized the numerical results in the comparison table using the values already obtained from Assignment 03, the theoretical exponential-EOS constants, the regressed exponential-EOS constants, and the NIST benchmark data. Overall, this assignment showed how an equation of state can be analyzed from both a theoretical and numerical point of view. It also demonstrated how critical-point constraints, regression methods, and residual-property theory can be combined in Excel to evaluate the performance of a proposed thermodynamic model.